Assignment - 01

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COURSE NAME: Cryptography & Network Security

90/100

1) *ENCRYT : "DEFEND THE WALL"

FORMULA:

$C = (P+3) \bmod 26$ (C : Ceasar Cipher with key = 3)

Shift :

D→G E→H F→I E→H N→Q D→G

T→W H→K E→H

W→Z A→D L→O L→O

Ciphertext : GHIKHG ZHK ZDOO

*DECRYPT "KHOOR ZRUOG" with key = 3

FORMULA:

$P = (C-3) \bmod 26$.

Shift

K→H H→E O→L, O→L, R→O.

Z→W, R→O U→R O→L G→D

*To find the key for SECRET→ VHFUHW and decrypt "WKLV LV DWVH

Key : S shifted to V. key = 3

Decrypt : THIS IS A TEST.

2) *ENCRYPT "FIRE"

In QWERTY substitution key

F → Y

I → O

R → K

E → T

Ciphertext : YOKT

* To find plain text

Ceaser cipher of shift - 3.

$P = (C+3) \bmod 26$

QEB → THE

NRFZH → QUICK.

YOLTR → BROWN

UJHHID → XMKKLG.

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3) * $P = (C-K) \bmod 26$

$Q - K = G$ $N - E = J$ $X - Y = Z$ $E - K = U$ $P - E = L$

∴ Plaintext : GJZULMCWGJCW

* Ciphertext : LXFOPVEFRNHR

Key : LEMON

$P = (C-K) \bmod 26$

Plaintext : ATTACKATDAWN

* Plaintext : Attackat Palace

Key : LEMON

$C = (P+K) \bmod 26.$

Ciphertext : LXFOPVEFDNWEOS

4) Encrypt "HI" $K = \begin{pmatrix} 3 & 3 \\ 2 & 5 \end{pmatrix}$

* vector Conversion : H = 7 I = 8

$$C = K.P \bmod 26 = \begin{pmatrix} 3 & 3 \\ 2 & 5 \end{pmatrix} \begin{pmatrix} 7 \\ 8 \end{pmatrix}$$

$$= \begin{pmatrix} 45 \\ 54 \end{pmatrix}$$

$45 \bmod 26 = 19$

$54 \bmod 26 = 2$

Ciphertext : TC

* Encrypt "OA" to "QC"

$$P = \begin{pmatrix} 14 \\ 0 \end{pmatrix}$$

$$C = \begin{pmatrix} 3 & 3 \\ 2 & 5 \end{pmatrix} \begin{pmatrix} 14 \\ 0 \end{pmatrix} = \begin{pmatrix} 42 \\ 28 \end{pmatrix}.$$

$42 \bmod 26 = 16 \rightarrow Q$

$28 \bmod 26 = 2 \rightarrow C$

Result : QC.

5) *Encrypt MEETMEAFTELUNCH" with "ZEBRAS"

6	3	2	4	1	5
Z	E	B	R	A	S
M	E	E	T	M	E
A	F	T	E	L	U
N	C	H	X	X	X

key.

Ciphertext : MLXETHEFCTEXEUXMAN

* Decrypt "EANMTEFCUHETREL" with "BOARD".

Matrix:

2	4	1	5	3	order
B	O	A	R	D	
M	H	E	R	F	
T	E	A	E	C	
E	T	N	L	U	

plaintext: MHERFTEAECETNLU

b) * Encrypt "HELLO" using one-Time pad

$C = (P+K) \bmod 26.$

$H + X = 30 \bmod 26 = 4 \rightarrow E$

$E + M = 16 \bmod 26 = 16 \rightarrow Q$

$L + C = 13 \bmod 26 = 13 \rightarrow N$

$L + K = 21 \bmod 26 = 21 \rightarrow V$

$L + O = 25 \bmod 26 = 25 \rightarrow Z$

Ciphertext: EQNVZ

* under ideal Conditions, one time pad achieves perfect secrecy because the key is Completely random, as long as the message, and never reused.
* Because the key is entirely uniform and random, a given ciphertext is equally likely to decrypt into any possible plaintext string of that length.

"WE ARE DISCOVERED" with "ZEBRA"

5 3 2 4 1
Z E B R A
W E A R E
D I . S C O V
V E R E D

Ciphertext : EO DASREIE ACEWDV

* Decrypt "EVLNEACDTKESEABROFDJDEECUWWN" with "TRIAL"

Key order A I L R T

28 characters is divided by 5 columns gives 5, 6, 5, 6, 6

plaintext : DEAR LAWYERS WE ARE DISCOVERED

* INFORMATION SECURITY with HACK.

3 1 2 4
H A C K
I N F O
R M A T
I O N S
E C U R
I T Y X

Ciphertext : NMCOTFANUYIRIEIOTSRx

8) MONO ALPHABETIC SUBSTITUTION:

Cephertext : UZQSOVUOHXMOFVGPOZPEVSGZW
ZOPFPESXUZUETVCG

Frequency:

Q → 5 Z → 5 P → 4 U → 4 V → 3

X → 3 E → 3 S → 3

Partial decryption:

Mapping Z → t, V → d, Q → s, U → l, Q → w, S → a

Plaintext: it was discovered yesterday that several informal

9) Monoalphabetic substitution with plaintext

Ciphertext: BJJW QJW YMJM NXF XNRUQWX

P = (C - 5) mod 26.

BJJW → WEER

QJW → PER

YMJW → THER

NX F XNRUQWX → IS A SIMPLE S

10) Vigenère cipher analysis

P = (C - "MATH" mod 26):

Z - M = 25 - 12 = N

I - A = I

C - T = J

V - H = O

Plaintext: NIJOS...